
Table of Contents

Preface.....	xiii
1. Getting Started.....	1
Snowflake Web User Interfaces	3
Prep Work	3
Snowsight Orientation	6
Snowsight Preferences	6
Navigating Snowsight Worksheets	8
Context Setting	8
Improved Productivity	14
Snowflake Community	17
Snowflake Certifications	19
Snowday and Snowflake Summit Events	19
Important Caveats About Code Examples in the Book	19
Code Cleanup	21
Summary	21
Knowledge Check	22
2. Creating and Managing the Snowflake Architecture.....	23
Prep Work	23
Traditional Data Platform Architectures	24
Shared-Disk (Scalable) Architecture	24
Shared-Nothing (Scalable) Architecture	25
NoSQL Alternatives	25
The Snowflake Architecture	26
The Cloud Services Layer	27
Managing the Cloud Services Layer	28
Billing for the Cloud Services Layer	28
The Query Processing (Virtual Warehouse) Compute Layer	29

Virtual Warehouse Size	30
Scaling Up a Virtual Warehouse to Process Large Data Volumes and Complex Queries	31
Scaling Out with Multiclustervirtual Warehouses to Maximize Concurrency	35
Creating and Using Virtual Warehouses	38
Separation of Workloads and Workload Management	42
Billing for the Virtual Warehouse Layer	44
Centralized (Hybrid Columnar) Database Storage Layer	45
Introduction to Zero-Copy Cloning	46
Introduction to Time Travel	46
Billing for the Storage Layer	46
Snowflake Caching	46
Query Result Cache	47
Metadata Cache	48
Virtual Warehouse Local Disk Cache	49
Code Cleanup	50
Summary	50
Knowledge Check	51
3. Creating and Managing Snowflake Securable Database Objects.	53
Prep Work	54
Creating and Managing Snowflake Databases	54
Creating and Managing Snowflake Schemas	63
INFORMATION_SCHEMA	65
ACCOUNT_USAGE Schema	69
Schema Object Hierarchy	70
Introduction to Snowflake Tables	70
Creating and Managing Views	76
Introduction to Snowflake Stages: File Format Included	79
Extending SQL with Stored Procedures and UDFs	82
User-Defined Function (UDF): Task Included	84
Secure SQL UDTF That Returns Tabular Value (Market Basket Analysis Example)	87
Stored Procedures	89
Introduction to Pipes, Streams, and Sequences	95
Snowflake Streams (Deep Dive)	96
Snowflake Tasks (Deep Dive)	102
Code Cleanup	108
Summary	108
Knowledge Check	109

4. Exploring Snowflake SQL Commands, Data Types, and Functions.	111
Prep Work	112
Working with SQL Commands in Snowflake	112
DDL Commands	113
DCL Commands	113
DML Commands	114
TCL Commands	114
DQL Command	114
SQL Query Development, Syntax, and Operators in Snowflake	115
SQL Development and Management	115
Query Syntax	117
Query Operators	124
Long-Running Queries, and Query Performance and Optimization	125
Snowflake Query Limits	126
Introduction to Data Types Supported by Snowflake	126
Numeric Data Types	127
String and Binary Data Types	129
Date and Time Input/Output Data Types	130
Semi-Structured Data Types	131
Unstructured Data Types	137
How Snowflake Supports Unstructured Data Use	138
Snowflake SQL Functions and Session Variables	141
Using System-Defined (Built-In) Functions	141
Creating SQL and JavaScript UDFs and Using Session Variables	144
External Functions	144
Code Cleanup	145
Summary	145
Knowledge Check	146
5. Leveraging Snowflake Access Controls.	147
Prep Work	148
Creating Snowflake Objects	150
Snowflake System-Defined Roles	154
Creating Custom Roles	155
Functional-Level Business and IT Roles	157
System-Level Service Account and Object Access Roles	158
Role Hierarchy Assignments: Assigning Roles to Other Roles	159
Granting Privileges to Roles	162
Assigning Roles to Users	165
Testing and Validating Our Work	166
User Management	169
Role Management	176
Snowflake Multi-Account Strategy	178

Managing Users and Groups with SCIM	178
Code Cleanup	179
Summary	180
Knowledge Check	180
6. Data Loading and Unloading.....	183
Prep Work	184
Basics of Data Loading and Unloading	184
Data Types	185
File Formats	185
Data File Compression	189
Frequency of Data Processing	189
Snowflake Stage References	190
Data Sources	191
Data Loading Tools	192
Snowflake Worksheet SQL Using INSERT INTO and INSERT ALL	
Commands	192
Web UI Load Data Wizard	204
SnowSQL CLI SQL PUT and COPY INTO Commands	210
Data Pipelines	212
Third-Party ETL and ELT Tools	221
Alternatives to Loading Data	222
Tools to Unload Data	222
Data Loading Best Practices for Snowflake Data Engineers	223
Select the Right Data Loading Tool and Consider the Appropriate Data	
Type Options	224
Avoid Row-by-Row Data Processing	224
Choose the Right Snowflake Virtual Warehouse Size and Split Files as	
Needed	225
Transform Data in Steps and Use Transient Tables for Intermediate Results	225
Code Cleanup	225
Summary	225
Knowledge Check	226
7. Implementing Data Governance, Account Security, and Data Protection and Recovery. .	227
Prep Work	228
Snowflake Security	230
Controlling Account Access	231
Monitoring Activity with the Snowflake ACCESS_HISTORY Account	
Usage View	235
Data Protection and Recovery	237
Replication and Failover	243
Democratizing Data with Data Governance Controls	244

INFORMATION_SCHEMA Data Dictionary	245
Object Tagging	245
Classification	249
Data Masking	251
Row Access Policies and Row-Level Security	253
External Tokenization	256
Secure Views and UDFs	257
Object Dependencies	257
Code Cleanup	258
Summary	258
Knowledge Check	258
8. Managing Snowflake Account Costs.....	261
Prep Work	262
Snowflake Monthly Bill	262
Storage Fees	264
Data Transfer Costs	264
Compute Credits Consumed	265
Creating Resource Monitors to Manage Virtual Warehouse Usage and	
Reduce Costs	266
Resource Monitor Credit Quota	268
Resource Monitor Credit Usage	268
Resource Monitor Notifications and Other Actions	269
Resource Monitor Rules for Assignments	269
DDL Commands for Creating and Managing Resource Monitors	271
Using Object Tagging for Cost Centers	276
Querying the ACCOUNT_USAGE View	276
Using BI Partner Dashboards to Monitor Snowflake Usage and Costs	277
Snowflake Agile Software Delivery	278
Why Do We Need DevOps?	278
Continuous Data Integration, Continuous Delivery, and	
Continuous Deployment	279
What Is Database Change Management?	280
How Zero-Copy Cloning Can Be Used to Support Dev/Test Environments	281
Code Cleanup	284
Summary	284
Knowledge Check	285
9. Analyzing and Improving Snowflake Query Performance.....	287
Prep Work	287
Analyzing Query Performance	287
QUERY_HISTORY Profiling	288
HASH() Function	288

Web UI History	289
Understanding Snowflake Micro-Partitions and Data Clustering	291
Partitions Explained	291
Snowflake Micro-Partitions Explained	294
Snowflake Data Clustering Explained	299
Clustering Width and Depth	299
Choosing a Clustering Key	303
Creating a Clustering Key	305
Reclustering	306
Performance Benefits of Materialized Views	306
Exploring Other Query Optimization Techniques	308
Search Optimization Service	308
Query Optimization Techniques Compared	309
Summary	310
Code Cleanup	310
Knowledge Check	310
10. Configuring and Managing Secure Data Sharing.....	313
Snowflake Architecture Data Sharing Support	314
The Power of Snowgrid	314
Data Sharing Use Cases	314
Snowflake Support for Unified ID 2.0	315
Snowflake Secure Data Sharing Approaches	316
Prep Work	317
Snowflake's Direct Secure Data Sharing Approach	318
Creating Outbound Shares	318
How Inbound Shares Are Used by Snowflake Data Consumers	330
How to List and Shop on the Public Snowflake Marketplace	333
Snowflake Marketplace for Providers	335
Standard Versus Personalized Data Listings	337
Harnessing the Power of a Snowflake Private Data Exchange	340
Snowflake Data Clean Rooms	341
Important Design, Security, and Performance Considerations	342
Share Design Considerations	342
Share Security Considerations	343
Share Performance Considerations	343
Difference Between Database Sharing and Database Cloning	343
Data Shares and Time Travel Considerations	344
Sharing of Data Shares	344
Summary	344
Code Cleanup	345
Knowledge Check	345

11. Visualizing Data in Snowsight.....	347
Prep Work	347
Data Sampling in Snowsight	348
Fixed-Size Sampling Based on a Specific Number of Rows	349
Fraction-Based Sampling Based on Probability	349
Previewing Fields and Data	349
Sampling Examples	352
Using Automatic Statistics and Interactive Results	353
Snowsight Dashboard Visualization	358
Creating a Dashboard and Tiles	358
Working with Chart Visualizations	361
Aggregating and Bucketing Data	363
Editing and Deleting Tiles	366
Collaboration	367
Sharing Your Query Results	368
Using a Private Link to Collaborate on Dashboards	368
Summary	369
Code Cleanup	369
Knowledge Check	370
12. Workloads for the Snowflake Data Cloud.....	371
Prep Work	372
Data Engineering	372
Data Warehousing	374
Data Vault 2.0 Modeling	374
Transforming Data within Snowflake	377
Data Lake	377
Data Collaboration	378
Data Monetization	379
Regulatory and Compliance Requirements for Data Sharing	379
Data Analytics	380
Advanced Analytics for the Finance Industry	380
Advanced Analytics for the Healthcare Industry	381
Advanced Analytics for the Manufacturing Industry and Logistics Services	382
Marketing Analytics for Retail Verticals and the Communications and Media Industry	382
Data Applications	383
Data Science	385
Snowpark	385
Streamlit	388
Cybersecurity Using Snowflake as a Security Data Lake	388
Overcoming the Challenges of a SIEM-Only Architecture	389
Search Optimization Service Versus Clustering	392

Unistore	401
Transactional Workload Versus Analytical Workload	401
Hybrid Tables	402
Summary	403
Code Cleanup	403
Knowledge Check	404
A. Answers to the Knowledge Check Questions.	405
B. Snowflake Object Naming Best Practices.	423
C. Setting Up a Snowflake Trial Account.	427
Index.	431